Solving by substitution

To use this method, rearrange one of the two equations so that the two unknown values (variables) are on different sides of the equation, then substitute this rearranged equation into the other equation. The new, combined equation contains only one unknown value and can be solved. Substituting the new value into one of the equations means that the other variable can also be found. Equations that cannot be solved by elimination can usually be solved by substitution.

▷ **Equation pair**
Solve this pair of simultaneous equations using the substitution method.

$$x + 2y = 7$$
$$4x - 3y = 6$$

Choose one of the equations, and rearrange it so that one of the two unknown values is the subject. Here x is made the subject by subtracting 2y from both sides of the equation.

choose one of the equations; this is the first equation

$$x + 2y = 7$$

make x the subject by subtracting 2y from both sides of the equation

$$x = 7 - 2y$$

2y must be subtracted from both sides of the equation

Then substitute the expression that has been found for that variable ($x = 7 - 2y$) into the other equation. This gives only one unknown value in the newly compiled equation. Rearrange this new equation to isolate y and find its value.

substitute the expression for x which has been found in the previous step

$$4x - 3y = 6$$

take the other equation

$$4(7 - 2y) - 3y = 6$$

this equation now has only one unknown value so it can be solved

$$28 - 8y - 3y = 6$$

multiply out the parentheses above: $4 \times 7 = 28$ and $4 \times -2y = -8y$

$$28 - 11y = 6$$

simplify the two y terms: $-8y - 3y = -11y$

$$-11y = -22$$

isolate the y term by subtracting 28 from this side

28 must also be subtracted from this side: $6 - 28 = -22$

$$\frac{-11y}{-11} = \frac{-22}{-11}$$

divide this side by −11 to isolate y ($-11y \div -11 = y$)

this side must also be divided by −11

$$y = 2$$

this is the value of y

Substitute the value of y that has just been found into either of the original pair of equations. Rearrange this equation to isolate x and find its value.

choose one of the equations; this is the first one

$$x + 2y = 7$$

$$x + (2 \times 2) = 7$$

because $y = 2$, 2y is $2 \times 2 = 4$

$$x + 4 = 7$$

work out the terms in the parentheses: $2 \times 2 = 4$

$$x = 3$$

subtract 4 from this side to isolate x

4 has been subtracted from the other side of the equation, so it must also be subtracted from this side: $7 - 4 = 3$

Both unknown variables have now been found—these are the solutions to the original pair of equations.

$$x = 3 \qquad y = 2$$